

Jonathan Coulter

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Education

Master of Science in Computer Science

University of Pittsburgh

Jan 2026 - Dec 2026

3.88 / 4.0

- Relevant Coursework: Algorithm Design, Natural Language Processing, Human-Computer Interaction, Interface Design

Bachelor of Science in Computer Science

University of Pittsburgh – Minors in Mathematics & Economics

Aug 2022 - Dec 2025

3.97 / 4.0

- Relevant Coursework: Data Structures & Algorithms, Web Development, Software Quality Assurance, Operating Systems

Experience

Microsoft

Software Engineer Intern

May 2026 – Present

Redmond, WA

- Azure Storage

BNY Mellon

Software Engineer Intern

Jun 2025 – Aug 2025

Pittsburgh, PA

- Built a unified monitoring system for a \$3 trillion daily transaction platform, automating 5+ manual operational workflows
- Integrated a custom AI SQL assistant using internal tools, allowing non-technical staff to retrieve and summarize data 70% faster
- Optimized the Spring Boot backend API using dynamic SQL query building, achieving a 90% reduction in query latency

Carnegie Mellon University

Software Engineer Intern

May 2024 – Dec 2024

Pittsburgh, PA

- Integrated cloud NoSQL Firebase into a Flask API, enabling modular deployment and cutting 30+ minutes of downtime per study
- Built a responsive UI with Bootstrap and jQuery for an AI-moderated chat platform, enhancing platform usability and engagement

Center for Analytical Approaches to Social Innovation

Data Science Research Fellow

Aug 2023 – Apr 2024

Pittsburgh, PA

- Leveraged Python and pandas to structure data involving the fragmented Allegheny County police-citizen complaint process
- Integrated data from 130+ municipalities into a GIS map, empowering local residents with actionable police accountability data

Projects

🔗 [whenlymeet.com](#) | *Flask, React, Google Calendar API, PostgreSQL, Docker, NGINX*

- Launched a full-stack group scheduling web app that attracted 300+ users, enabling creation and sharing of availability polls
- Integrated Google Calendar API to autofill availability, powering 1000+ availability responses with minimal user input
- Designed and optimized a mobile-first React UI, driving 250+ events created and seamless scheduling across devices

Echo | *Flask, Spotify API, OpenCV, ML*

- Developed a full-stack platform integrated with the Spotify API to generate 20+ song playlists for the user based on their mood
- Implemented real-time facial recognition pipeline using OpenCV and a DeepFace machine learning model for emotion detection
- Built a REST API with Flask to handle user authentication, data processing, and third-party API integration

Leadership

Pitt Computer Science Club

Board Member

Apr 2025 – Present

Pittsburgh, PA

- Led the club's mock interview service, helping 20+ students prepare for technical + behavioral interviews, yielding a 80% offer rate
- Co-led Bit/Byte, a mentorship program connecting 100+ new students with mentors leading to 35% receiving internship offers

University of Pittsburgh

Teaching Assistant & Peer Tutor

Jan 2024 – Present

Pittsburgh, PA

- Led 25 students through weekly recitation discussions and technical exercises to clarify Data Structures & Algorithms concepts
- Hosted office hours available to 150+ students where I developed personalized explanations for diverse learning styles

Stanford University - Code in Place

Section Leader

Apr 2025 – May 2025

Remote

- Led a weekly virtual section of 15 students through an international summer session of Stanford's CS 106A on Python programming

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, C++, HTML/CSS, SQL, Rust, Dart, Haskell, Swift

Frameworks: Flask, React, Spring Boot, Flutter, Angular, Node.js, JUnit, jQuery

Tools: Git, Docker, Firebase, PostgreSQL, Google Cloud Platform, NGINX, Linux, Excel, Jupyter

Concepts: Artificial Intelligence, Agile, Scrum, CRUD, Object-Oriented Programming, Test-Driven Development, CI/CD, Cloud Computing